STUDENT ACADEMIC PROGRESSION & PORTFOLIO MANAGEMENT SYSTEM

Comprehensive DBMS Technical Architecture & Engineering Report • Relational 3NF Schema • Cloud Deployed

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1. Abstract

The Student Academic Progression & Portfolio Management System is an enterprise-grade relational database management application designed to eliminate data silos in collegiate academic tracking. Implementing a rigorous Third Normal Form (3NF) relational schema, the system provides real-time computation of Semester Grade Point Average (SGPA) and progressive Cumulative Grade Point Average (CGPA) on a 10-point scale. To ensure high reliability, an asynchronous Outbox Pattern decouples local SQL commits from external Google Sheets synchronization, preventing UI latency while providing complete spreadsheet integration across 7 relational entities.

2. Problem Statement & Objectives

Traditional educational record systems suffer from disparate spreadsheets, lack of referential integrity, unverified manual CGPA calculations, and absence of extracurricular tracking. Key project objectives include: (1) Enforcing ACID compliance and relational integrity via foreign keys with cascading deletions; (2) Dynamic recalculation of progressive CGPA upon any mark update; (3) Durable background synchronization with Google Sheets; and (4) Providing automated multi-sheet Excel reporting.

3. Relational Schema & Third Normal Form (3NF)

The database design conforms to Third Normal Form, ensuring that every non-key attribute is non-transitively dependent on the primary key. Seven core tables model academic life, supported by dedicated sync tables:

Table Name	Primary Key	Foreign Keys	Key Constraints / Purpose
students	id (PK)	None	student_code UNIQUE, email UNIQUE
school_records	id (PK)	student_id -> students.id	ON DELETE CASCADE, percentage auto-calculated
semesters	id (PK)	student_id -> students.id	UNIQUE(student_id, semester_number), ON DELETE CASCADE
student_marks	id (PK)	semester_id -> semesters.id	UNIQUE(semester_id, subject_code), 10-pt grade GP
gap_years	id (PK)	student_id -> students.id	ON DELETE CASCADE, official justifications
seminars	id (PK)	student_id -> students.id	ON DELETE CASCADE, conference & participation role
activities	id (PK)	student_id -> students.id	ON DELETE CASCADE, technical/sports achievements
sync_outbox	id (PK)	None	Durable queue: status (PENDING/SYNCED/FAILED)
sync_log	id (PK)	None	Audit trail with timestamp, operation, error logs

4. Academic SGPA & Cumulative CGPA Calculation Engine

Grading Standard: Marks are converted to letter grades and Grade Points (GP): ≥90% → 'O' (10.0), 80-89% → 'A+' (9.0), 70-79% → 'A' (8.0), 60-69% → 'B+' (7.0), 55-59% → 'B' (6.0), 50-54% → 'C' (5.0), 40-49% → 'P' (4.0), <40% → 'F' (0.0).
Semester SGPA Formula: SGPA = ∑(Credits × GradePoint) / ∑(Credits)
Cumulative CGPA Formula: $CGPA_k = \sum_{i=1..k}(\text{Semester Credits}_i \times SGPA_i) / \sum_{i=1..k}(\text{Semester Credits}_i)$

5. Google Sheets Integration & Outbox Architecture

External synchronization is orchestrated via an asynchronous Outbox Pattern. When a user submits an update, the SQL transaction commits first. A payload is atomically saved into sync_outbox. The Google Apps Script webhook receives updates, validates the secret token, and executes idempotent upserts matching record primary keys across all 7 target worksheets. This completely isolates the end-user application from Google Sheets rate-limiting or temporary API downtimes.

6. Automated Verification & Testing Results

An automated test suite using pytest validates every core system layer. 100% of tests passed:

Test Case Name	Module Verified	Status	Details
test_health_check	app.py (/health)	PASSED	Verified 200 OK, database alive, UTC timestamp
test_dashboard_loads	templates/dashboard.html	PASSED	Verified aggregate KPI queries & Chart.js
test_grade_and_sgpa_calculations	services/calculations.py	PASSED	Verified 10-point grade conversions & SGPA = 9.20
test_cgpa_progressive_recalculation	services/calculations.py	PASSED	Verified multi-semester cumulative weighted CGPA
test_student_directory_and_search	app.py (/students)	PASSED	Verified parameterized search and empty state handling
test_student_crud	services/db.py & outbox	PASSED	Verified INSERT, UPDATE, outbox queueing, and DELETE
test_semester_and_mark_cascade	services/db.py	PASSED	Verified mark addition auto-updates semester SGPA & credits
test_excel_export	services/export_excel.py	PASSED	Verified openpyxl generation with all 7 worksheets

7. Cloud Deployment & Security Architecture

Cloud Deployment: Container-ready with Gunicorn (gunicorn app:app --workers 2) configured for Render / Heroku with Procfile and render.yaml.
Security: 100% of database interactions utilize parameterized queries preventing SQL injection; all sensitive credentials are isolated in environment variables; CSRF and double-submit guards are enforced on frontend forms.